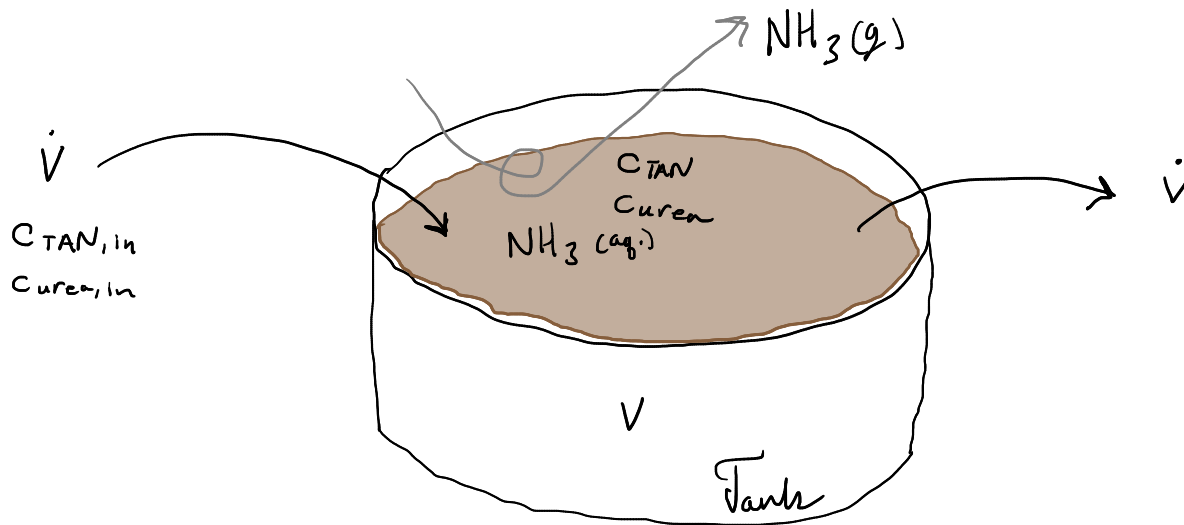


More details on the conceptual and mathematical models for the ammonia emission problem

A tank open to the air receives manure at a constant volumetric flow rate and it is removed at the same constant rate. The fresh manure has a fixed urea and total ammoniacal nitrogen (TAN) concentration. Urea can hydrolyze to ammonia (although it does it rather quickly without an inhibitor) and ammonia can volatilize from the tank (although only the uncharged NH_3 species is volatile).



$\dot{V}$ = constant manure flow in and out

$$\text{m}^3 \text{ s}^{-1}$$

V = constant manure volume in tank

$$\text{m}^3$$

C_{TAN} = total ammoniacal N concentration

$$\text{kg m}^{-3}$$

C_{urea} = urea concentrations as N

$$\text{kg m}^{-3}$$

mass balance:

TAN accumulation = TAN pumping in - NH_3 volatilization out + TAN generation - TAN pumping out

Urea accumulation = Urea pumping in - urea hydrolysis - urea pumping out

The dynmod() function uses a mass balance normalized by manure volume V . So units are $\text{kg/m}^3\text{-s}$ (kg of N) for each term.

23 & 27 Oct 2025. Correction to NH_3 volatilization below (was $\frac{A}{V}$ now $\frac{A}{d}$ because $\frac{A}{V} = \frac{1}{d}$.)
In NH_3 volatilization equation below, d = manure depth (m). And A = surface area (m^2),
 V = manure volume (m^3).

The constitutive equations are given below. These also have units of $\text{kg/m}^3\text{-s}$ because of the normalization by manure volume. This is also by we have retention time and not flow below.

TAN pumping in = $C_{\text{TAN},in} / \tau$ ----- Retention time = $\frac{V}{\dot{V}}$ (τ is "tau")

TAN pumping out = C_{TAN} / τ ----- Tank conc.

27 Oct 2025. Correction. H is aq:g so should be multiplied by gas phase conc.

NH_3 volatilization out = $k_L / d \cdot (f_{\text{NH}_3} \cdot C_{\text{TAN}} - C_{\text{NH}_3,g} \cdot H)$
(gas-phase) conc. (kg m^{-3})

TAN generation = $\alpha \cdot C_{\text{urea}}$

Urea pumping in = $C_{\text{urea},in} / \tau$

Urea hydrolysis = $\alpha \cdot C_{\text{urea}}$

Urea pumping out = C_{urea} / τ ----- First-order rate constant (s^{-1})